

FINAL UAV DRONE PROJECT METHODOLOGY

Human Detection → Tracking → Pedestrian Attribute Recognition

Crisp, print-ready, detailed consolidated version

Final architecture

YOUR 30-MIN / 16-GB DRONE VIDEO → frame sampling → RT-DETRv2 human detector → corrected target-domain boxes → ByteTrack → person IDs → K diverse observations/person → Swin + DFSM → activation/confidence evidence → S-ACRM → K-frame fusion → final attributes.

Roles: RT-DETRv2 finds people; ByteTrack links detections into identities; Swin learns attribute features; DFSM encourages diverse feature discovery; S-ACRM relates activation evidence and confidence; K-frame fusion combines observations; your drone data performs target-domain adaptation.

This version intentionally removes the previous time-estimate table and 8-week schedule. It consolidates the detailed content from the previous final PDF with the new two-methodology material, while tightening margins, spacing and typography for printing.

Core research hypothesis: multiple diverse UAV observations of the same person can provide more reliable attribute evidence than one observation, particularly under occlusion, viewpoint change and small-object conditions.

1. EXACT PROJECT SYSTEM

Component	Exact job	Output
RT-DETRv2	Detect humans in each drone frame	person boxes + confidence
ByteTrack	Associate detections over time	person_id + trajectory
Swin Transformer	Extract person/attribute features	feature representation + logits
DFSM	Suppress dominant/random regions to discover complementary features	more diverse features
S-ACRM	Relate activation energy and attribute confidence	attribute relation matrix
K-frame fusion	Combine multiple observations of the same ID	final relation representation
Final classifier	Convert fused evidence to attributes	multi-label attributes

End-to-end operation

Video → frames → detector → boxes → ByteTrack → same-person sequence → person crops → Swin/DFSM → Vc + Ve → S-ACRM → R1...RK → fusion → final attributes.

Do not collapse these roles: detection is not tracking; tracking is not attribute recognition; detector predictions are not attribute labels; automatic boxes are not final ground truth.

2. DATASETS — USE EACH FOR A SPECIFIC JOB

Dataset	Use in project	Priority
VisDrone	Real UAV detection; optional UAV tracking validation	Essential for detector
UAV-Human	UAV pedestrian attributes, video/person sequences	Essential for attributes
Your drone video	Target-domain detector/attribute adaptation + final test	Mandatory
PA-100K	General PAR pretraining/benchmark	Optional
PETA	General multi-view PAR benchmark	Optional
RAPv1/RAPv2	Additional PAR benchmarks	Optional
UPAR	Unified PAR pretraining across datasets	Optional
SkyScenes	Synthetic aerial pretraining; convert instance masks to boxes	Optional/strong

2.1 VisDrone

Use **VisDrone-DET** for human detection training/validation. Use **VisDrone-VID/MOT** only if you want an additional tracking experiment. Keep pedestrian as the target human class if the final detector is one-class person detection.

Official: <https://github.com/VisDrone/VisDrone-Dataset>

2.2 UAV-Human

Use for UAV-specific pedestrian attributes and person-level sequences. The reported attribute taxonomy includes gender, hat, backpack, upper-clothing colour/style and lower-clothing colour/style. The UAV viewpoint makes it substantially more relevant to your target problem than relying only on general pedestrian datasets.

Official GitHub: <https://github.com/sutdcv/UAV-Human>

Dataset page: <https://sutdcv.github.io/uav-human-web/>

2.3 PA-100K / PETA / RAP / UPAR

Use these only when they improve pretraining or benchmarking. PA-100K provides 100,000 pedestrian images and 26 binary attributes. PETA provides 19,000 images of 8,705 pedestrians with many attributes/viewpoints. RAPv1/RAPv2 are optional additional PAR benchmarks. UPAR harmonizes attributes across multiple datasets.

PA-100K: <https://github.com/xh-liu/HydraPlus-Net> | PETA: <https://mmlab.ie.cuhk.edu.hk/projects/PETA.html>

RAP: <https://github.com/valencebond/RAP-dataset> | UPAR: https://github.com/speckean/upar_dataset

2.4 Your own drone data

Public datasets initialize/adapt the model; your drone video is the **target domain**. Always keep a held-out portion of your own video for the final test.

3. STEP 1 — VIDEO AUDIT AND FRAME SAMPLING

Your 30-minute / 16-GB video may contain roughly 54,000 frames at 30 FPS. Do not start by extracting and annotating every frame.

Audit

- FPS and resolution.
- Camera altitude and view angle.
- Typical person pixel height/width.
- Motion blur and camera movement.
- Lighting/weather variation.
- Crowd density and occlusion.
- Scene changes and repeated backgrounds.

Sampling

Start at **1 frame / 2 s $\approx$ 900 frames**. If scenes change quickly, use **1 frame / s $\approx$ 1,800 frames**. Add difficult/rare frames after visual inspection and remove near-duplicates.

Sampling logic

Uniform coverage $\rightarrow$ inspect $\rightarrow$ add hard cases $\rightarrow$ remove duplicates $\rightarrow$ annotation set $\rightarrow$ segment-level train/val/test split.

Critical: do not randomly distribute adjacent video frames across train/test. Use disjoint video segments and/or identities to prevent leakage.

4. STEP 2 — PRETRAINED DETECTOR AND AUTOMATIC PRE-ANNOTATION

Recommended main detector: **RT-DETRv2**. Run the pretrained model first to establish the baseline on your drone footage. Hugging Face Transformers provides RT-DETRv2 documentation and fine-tuning support.

RT-DETRv2 docs: https://huggingface.co/docs/transformers/model_doc/rt_detr_v2

Automatic annotation

Pretrained RT-DETRv2 $\rightarrow$ predicted person boxes $\rightarrow$ CVAT $\rightarrow$ human correction $\rightarrow$ final ground truth.

For each detection save frame_id, class, confidence, x1, y1, x2, y2. Use a permissive threshold for pre-annotation so weak candidates can be reviewed; choose the final inference threshold from validation data.

CVAT automatic annotation: <https://docs.cvat.ai/docs/annotation/auto-annotation/automatic-annotation/>

Human correction

- Add missed people.
- Delete false positives.
- Tighten/repair inaccurate boxes.
- Check tiny and overlapping people.
- Flag ambiguous frames instead of forcing unreliable labels.

Optional YOLO role

If you also want a fast annotation assistant/baseline, an Ultralytics COCO-pretrained YOLO detector can be used in exactly the same way: pretrained inference $\rightarrow$ person boxes $\rightarrow$ CVAT correction. It is **not** the final research detector if RT-DETRv2 is your locked architecture.

Ultralytics detection docs: <https://docs.ultralytics.com/tasks/detect>

Do not treat YOLO/RT-DETR automatic predictions as ground truth.

5. STEP 3 — OPTIONAL SKYSCENES AERIAL PRETRAINING

SkyScenes is synthetic UAV data with semantic/instance/depth annotations. It is **not simply a ready-made YOLO box dataset**. Use its pedestrian instance annotations to derive person boxes.

Links: [official](#) | [Hugging Face](#) | [paper](#)

Exact conversion

- Select heights/pitches/weather closest to your flight.
- Select pedestrian instances.
- For each instance mask, obtain xmin, xmax, ymin, ymax.
- Convert to YOLO normalized coordinates.
- Overlay random samples to verify boxes.
- Remove invalid/empty/tiny unusable instances.
- Train aerial detector from COCO-pretrained weights.
- Continue to VisDrone real-UAV data.
- Finally adapt to your own drone boxes.

YOLO box conversion

class_id=0; x_center=(xmin+xmax)/(2W); y_center=(ymin+ymax)/(2H); width=(xmax-xmin)/W; height=(ymax-ymin)/H

Recommended detector ladder

COCO pretrained $\rightarrow$ SkyScenes aerial $\rightarrow$ VisDrone real UAV $\rightarrow$ your drone target domain

SkyScenes is an optional addition. Keep it only if the validation experiment demonstrates a useful improvement relative to the extra preprocessing.

6. STEP 4 — VISDRONE + TARGET-DOMAIN DETECTOR FINE-TUNING

Prepare VisDrone-DET by mapping the pedestrian category to your single target class **person**. Convert annotations to the detector's required format if necessary.

Train/evaluate the following controlled detector ladder:

Model	Training data	Purpose
D0	COCO-pretrained RT-DETRv2	General baseline
D1	D0 + VisDrone	Real-UAV adaptation
D2	D1 + your corrected drone frames	Target-domain adaptation
D3 optional	COCO → SkyScenes → VisDrone → your drone	Synthetic + real aerial adaptation

Keep the same held-out drone test segment for all detector variants. Report Precision, Recall, mAP@50, mAP@50:95 and FPS. Also stratify results by small-person size and occlusion if your labels permit.

Scientific meaning: if D0 is already strong, that is a valid baseline. Your contribution is demonstrated by measurable improvement in aerial/small/occluded/target-domain cases rather than by claiming the pretrained model cannot detect people.

7. STEP 5 — BYTE TRACKING AND PERSON SEQUENCES

Detector output → BvteTrack input

```
frame_id + [x1,y1,x2,y2,confidence,class=person]
```

BvteTrack output

```
frame_id + person_id + bbox + confidence
```

Official ByteTrack: <https://github.com/FoundationVision/ByteTrack>

Example: Person 17 → frame 100 crop 1 → frame 104 crop 2 → frame 108 crop 3 → frame 112 crop 4 → frame 116 crop 5.

Select **diverse** observations, not simply adjacent frames. Prefer changes in viewpoint, pose, visibility and lighting. Discard heavily blurred or duplicate crops.

Tracking is the bridge to the WACV multi-appearance idea: ByteTrack gives you the same-person identity required to combine multiple observations.

8. STEP 6 — BUILD THE UAV ATTRIBUTE DATASET

Start with UAV-Human attributes that are genuinely visible from your drone camera.

Recommended initial set

Backpack; hat; upper-clothing colour; upper-clothing style; lower-clothing colour; lower-clothing style. Add other UAV-Human attributes only when annotation is reliable.

Treat difficult/unobservable attributes as **unknown**, not automatically negative. Gender should be included only if the image quality/viewpoint supports defensible annotation.

Record per observation

```
person_id | frame_id | bbox | crop_path | attribute_1 ... attribute_M
```

For each person, initially annotate 5–10 representative crops. Split by identity or disjoint video segment; never put frames of the same identity into both train and test.

External attribute data

UAV-Human is the most relevant external source. PA-100K/PETA/RAP/UPAR can provide general PAR pretraining, but your own drone crops are necessary for final target-domain validation.

9. METHODOLOGY 1 — FAITHFUL REPRODUCTION + UAV ADAPTATION

This is the **safer baseline methodology**. It preserves the WACV paper's multi-observation/S-ACRM mechanism and adapts the detector and attribute pipeline to UAV data.

Exact pipeline

```
VisDrone → RT-DETRv2 → own-drone fine-tuning → ByteTrack → person sequences → attribute backbone → S-ACRM → multi-frame fusion → final attributes
```

Part A — Detector

Train/fine-tune RT-DETRv2 on VisDrone and corrected drone boxes. The detector answers **where is the person?**

Part B — Tracker

ByteTrack associates boxes over time. The tracker answers **which detections belong to the same person?**

Part C — Attribute baseline

For faithful reproduction, the uploaded WACV methodology uses a DeepMAR-style baseline with an ImageNet-pretrained ResNet50 backbone. For the UAV implementation, you may replace this feature extractor with Swin while keeping S-ACRM unchanged; if doing so, report it explicitly as a backbone modification.

Part D — S-ACRM

For each observation obtain attribute confidence V_c and class-activation energy V_e , then construct the $M \times M$ relation representation. The paper's mechanism is the source of this relation learning.

Part E — K-observation fusion

For the same person, compute $R_1 \dots R_K$ and use the attribute-wise maximum to form R_c . Compare $K=1$ against $K=2,3,4,5$.

Advantages

- Closest to the uploaded WACV methodology.
- Clear reproducibility baseline.
- Easy single-frame vs multi-frame comparison.
- Lower research risk.

Use Methodology 1 as the baseline even if Methodology 2 becomes the final contribution.

10. METHODOLOGY 2 — RECOMMENDED RESEARCH CONTRIBUTION

This is the stronger proposed system: UAV-specific detection/tracking + Swin/DFSM + S-ACRM + K-frame evidence fusion.

Exact pipeline

Aerial detector adaptation → ByteTrack → Swin + DFSM → activation/confidence evidence → S-ACRM → K-frame fusion → attributes

Stage 1 — UAV detection

Use VisDrone and your corrected drone boxes. Optionally insert SkyScenes before VisDrone.

Stage 2 — Tracking

ByteTrack creates persistent person IDs and diverse observations.

Stage 3 — Swin

Use Swin as the attribute feature extractor. Its shifted-window architecture provides local and longer-range feature modelling.

Stage 4 — DFSM

Add Peak Region Suppression (PRS) and Random Region Suppression (RRS). Start from the second uploaded paper's reported $\alpha=0.9$, $n=2$, $r=0.25$.

Stage 5 — S-ACRM

Use activation energy and confidence to construct the attribute relation representation.

Stage 6 — K-frame fusion

$K=1$ is the single-frame baseline; $K=2,3,4,5$ test multi-observation benefit. Use $K=5$ as the main proposed setting initially, because the uploaded WACV experiments reported $K=5$ as strongest among the tested K values in their multi-appearance experiments.

Why this is the research contribution

DFSM: diverse feature discovery within an image. **S-ACRM + K fusion**: evidence aggregation across observations. **Detector + ByteTrack**: provides the UAV-specific person sequence that makes the multi-observation stage possible.

11. SWIN + DFSM — EXACT IMPLEMENTATION

Single-frame baseline

Person crop → 224×224 → Swin → M attribute logits → sigmoid → M attribute probabilities

DFSM settings from uploaded paper

$\alpha=0.9$; $n=2$; $r=0.25$. Reported starting training setup: 224×224 , random flip/crop, SGD, batch 16, momentum 0.9, weight decay 0.0005, learning rate $1e-4$, 80 epochs.

Required ablation

Swin → Swin+PRS → Swin+RRS → Swin+PRS+RRS

Interpretation: this isolates whether peak-region suppression, random-region suppression or their combination produces the gain.

12. S-ACRM — EXACT IMPLEMENTATION

For observation i

- Run the attribute backbone.
- Obtain the attribute prediction.
- Obtain confidence vector V_c .
- Obtain class-activation energy vector V_e .
- Construct the $M \times M$ relation representation.

Concentual relation expression

$R_i = \text{softmax}(V_{e_i} \times V_{c_i})$

For K observations

$R_1, R_2, \dots, R_K \rightarrow \text{attribute-wise max} \rightarrow R_c \rightarrow \text{final fully connected classifier} \rightarrow \text{final attributes}$

The WACV methodology also uses BCE-style losses and weighted BCE to address attribute imbalance. For a faithful reproduction, follow the uploaded paper's exact loss definition.

Key idea: if a backpack is unclear in frame 1 but obvious in frame 4, frame 4 should contribute strong positive evidence rather than letting the weak frame dominate the final decision.

13. EXACT EXPERIMENTS AND ABLATIONS

ID	Experiment	What it proves
D1	Pretrained RT-DETRv2 vs fine-tuned RT-DETRv2	Target-domain adaptation benefit
D2	VisDrone vs VisDrone + own drone	Real-UAV + target-domain benefit
D3	Optional COCO→SkyScenes→VisDrone→drone	Synthetic aerial pretraining benefit
T1	Detector + ByteTrack	Identity continuity
A1	Swin only	Attribute baseline
A2	Swin+PRS / Swin+RRS / both	DFSM component contribution
A3	K=1,2,3,4,5	Multi-observation contribution
A4	Without vs with S-ACRM	Activation/confidence relation contribution
A5	Full model	Overall proposed system

Metrics

Detection: Precision, Recall, mAP@50, mAP@50:95, FPS. Tracking: IDF1, MOTA, ID switches. Attribute recognition: mA, Accuracy, Precision, Recall, F1.

Most important comparison: **K=1 vs K>1**. This directly tests whether the tracked multi-observation mechanism adds value beyond a strong single-frame attribute model.

14. EXACT FULL RUN ON YOUR 30-MINUTE VIDEO

Phase	Operation	Deliverable
1	Audit video + sample frames	Candidate frame set
2	Run pretrained RT-DETRv2	Baseline detections
3	Automatic annotation + CVAT correction	Ground-truth person boxes
4	Prepare VisDrone	Aerial detection dataset
5	Optional SkyScenes mask→box conversion	Synthetic aerial dataset
6	Fine-tune detector	Final detector checkpoint
7	Run detector + ByteTrack	Person trajectories
8	Extract diverse K observations	Per-person crop sequences
9	Prepare UAV-Human/general PAR data	Attribute training data
10	Train Swin baseline	Single-frame PAR model
11	Add DFSM + ablations	Diverse-feature results
12	Add S-ACRM + K fusion	Multi-observation model
13	Evaluate K=1...5	Ablation evidence
14	Run full video	Final video + CSV/JSON

No fixed project calendar is imposed here; execute each stage only after the previous stage's data/checkpoint has been validated.

15. FINAL OUTPUTS YOU SHOULD HAVE

- Final detector checkpoint.
- Held-out drone detection metrics.
- ByteTrack person-ID trajectories.
- Person-crop dataset with identity and frame metadata.
- Swin baseline results.
- DFSM ablation results.
- S-ACRM ablation results.
- K=1–5 multi-observation results.
- Final annotated 30-minute video.
- Per-person attribute CSV/JSON.
- Qualitative examples: tiny people, occlusion, viewpoint change, clear vs unclear attributes.

Recommended per-person output record

```
video_id | frame_id | person_id | x1 | y1 | x2 | y2 | detector_conf | backpack | hat | upper_color | upper_style | lower_color | lower_style | attribute_confidence
```

16. FINAL LOCKED METHODOLOGY

Use Methodology 1 as the faithful baseline and Methodology 2 as the main research contribution.

Baseline: VisDrone → RT-DETRv2 → own-drone fine-tuning → ByteTrack → person sequences → attribute backbone → S-ACRM → multi-frame fusion → attributes.

Proposed: aerial detector adaptation → ByteTrack → Swin + DFSM → S-ACRM → K-frame fusion → final attributes.

Final detector question: Where is the person?

Final tracker question: Which detections belong to the same person?

Attribute model question: What visual evidence is present?

Multi-observation question: Can evidence visible in one of several observations improve the final attribute decision?

This separation makes the project scientifically clean: public UAV datasets provide initialization, your corrected drone data provides target-domain supervision, ByteTrack supplies identity, and the two uploaded papers supply the attribute-learning mechanisms.

17. REFERENCES / OFFICIAL LINKS

Uploaded papers

[1] Thakare et al., *Let's Observe Them Over Time: An Improved Pedestrian Attribute Recognition*, WACV 2024 — basis for S-ACRM, activation/confidence evidence, multi-appearance fusion and PAR evaluation.

[2] Uploaded paper on *Diverse Feature Discovery Transformer (DFDT)* — basis for Swin, DFSM, PRS/RRS and reported training settings.

UAV / aerial datasets

VisDrone: [official repository](#)

UAV-Human: [official repository](#) | [dataset page](#)

SkyScenes: [project](#) | [Hugging Face](#) | [paper](#)

PAR datasets

PA-100K: [resources](#) | PETA: [official page](#) | RAP: [repository](#) | UPAR: [repository](#)

Models / tools

RT-DETRv2: [Hugging Face documentation](#)

ByteTrack: [official GitHub](#)

CVAT automatic annotation: [documentation](#)

Optional YOLO annotation/baseline: [Ultralytics detection docs](#)

FINAL REPORT — LINK VERIFICATION

Checked against current official web resources on 18 August 2026. The final PDF uses stable official repositories/documentation. SkyScenes and UAV-Human project-page links were redirected to their official GitHub repositories because those are directly reachable and maintained.

Resource	Destination	Check
VisDrone official repository	https://github.com/VisDrone/VisDrone-Dataset	LIVE
UAV-Human official repository	https://github.com/sutdcv/UAV-Human	LIVE
PA-100K / HydraPlus-Net	https://github.com/xh-liu/HydraPlus-Net	LIVE
PETA official page	https://mmlab.ie.cuhk.edu.hk/projects/PETA.html	LIVE
RAP dataset repository	https://github.com/valencebond/RAP-dataset	LIVE
UPAR official repository	https://github.com/speckean/upar_dataset	LIVE
Hugging Face RT-DETRv2 docs	https://huggingface.co/docs/transformers/model_doc/rt_detr_v2	LIVE
CVAT automatic annotation docs	https://docs.cvat.ai/docs/annotation/auto-annotation/automatic-annotation/	LIVE
Ultralytics detection docs	https://docs.ultralytics.com/tasks/detect	LIVE
SkyScenes official GitHub repository	https://github.com/hoffman-group/SkyScenes	LIVE
SkyScenes Hugging Face dataset	https://huggingface.co/datasets/hoffman-lab/SkyScenes	LIVE
SkyScenes paper (arXiv)	https://arxiv.org/abs/2312.06719	LIVE
ByteTrack official repository	https://github.com/FoundationVision/ByteTrack	LIVE

Access note: LIVE means the official resource/repository/documentation is reachable. Some datasets may still require forms, login, license acceptance, or research-only access for downloading files.